# MODEL QUESTION PAPER

## Physical Science

## Part-A and B

Class: **X** Max.Marks:4**0**

Time:1.35 hrs.

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**General Instructions:**

i. Read the question paper carefully and understand.
ii. Answer the questions under Part-A in the answer sheet provided
iii. Write the answer to the questions under Part-B in the space provided and attach it to the Part-A answer sheet.
iv. Part-A contains 3sections
v. Write the answers following the instructions given in the each section.

### PART-A

**Marks:30**

### Section – I

**3 X 2 = 6 Marks**

**Instructions:**

i. Answer any 3 of the following questions
ii. Write the answers in 1-2 sentences.
iii. Each question carries 2 marks

1. What happens if concave mirror is used as a rear view mirror?
2. In an experiment ***NaCl*** is reacts with ***$H_2SO_4$*** and evolves ***HCl*** gas. What is the result when we test the evolved gas with wet litmus paper?
3. The electronic configuration of sodium is $1s^2 2s^2 2p^6 3s^1$, and then writes the 4 quantum numbers of the first electron in K shell.
4. Mention the chemical equation of reaction between the Mg and $O_2$ and write its physical state of reactants and products.
5. A doctor suggested to use the +50 cm focal length lens to correct a person's eye defect, mention which type of defect he is suffering and mention type of lens doctor suggested.
6. Which material is used to make the filament of the bulb? Why?

## Section – II

**2 X 4 = 8 Marks**

**Instructions:**

i. Answer any 2 of the following questions
ii. Write the answers in 4-5 sentences.
iii. Each question carries 4 marks

7. Find the focus and position of the lens by drawing a ray diagram using the position of object A and the image $A^1$ in the given diagram.

8. What would happen if corrosion of metals is not prevented?
9. The elements belong to the modern periodic table the atomic numbers of the elements are 20, 26. Write the elements belong to which period and group.
10. Mention the day to day uses of the convex lenses.

## Section – III

**2 X 8 = 16 Marks**

**Instructions:**

i. Answer any 2 of the following questions
ii. Write the answers in 8 -12 sentences.
iii. Each question carries 8 marks

11. Mention the precautions to be taken in the experiment to show that hydrogen gas is evolved when metals react with acids. Mention the experimental procedure.

12. In the following table nature of some solutions are given.

| Model solution | NaOH | Saliva | Sea water | HCl | NaCl | Pure water | Acitic acid | Blood |
|---|---|---|---|---|---|---|---|---|
| $P^H$ value | **13** | **6** | **8** | **1** | **7** | **7** | **6** | **7.4** |

Answer the following questions.

a) Write the acids and bases.
b) In which of the above two material react to get neutralization reaction.
c) Mention the strong acids and weak bases from the above table.
d) Mention the nature of the product when NaOH reacts with acetic acid ?

13. Mention the main features of the modern periodic table.

14. 10V battery is connected in a electric circuit. Between A and B points 20 Ω and 30 Ω resisters are connected to get least resistance in the circuit. Find the resultant resistance between A and B and find current in the circuit.

# MODEL QUESTION PAPER

## Physical Science

## Part - B

Class: **X** Marks:10

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**Instructions:**

i. Answer all the questions

ii. Choose the correct answer from the given options and write the corresponding letter (A/B/C/D) in the capital letters in the space provided

iii. Each question carries 1 mark

iv. Marks will not be given for over writing/ re-writing / erased answers.

1. The focal length of a symmetrical converging lens is equal to the radius of curvature. Then the refractive index is ( )

   A) 1 B) 1.5 C) 2 D) 3

2. After white washing of house, walls will appear to be brighter after 1 or 2 days. The white substance formed on the walls is ( )

   A) $CaO$ B) $H_2O$ C) $Ca(OH)_2$ D) $CaCO_3$

3. When a honey bee stings and injected ethnoic acid, what is used to get relief ( )

   A) Calcium hydroxide B) Lemon water C) Common salt D) Soda water

4. If "–D" power lens is used for the correction of the eye defect, then the lens used is ( )

   A) Bi convex lens B) Plano concave lens

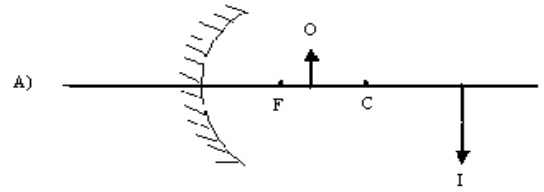
   C) Concave convex lens D) concave lens

5. 3*s*, 3*p*, 4*s*, 3*d* of these, more energy level orbital is ( )

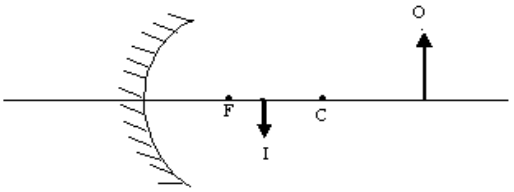
   A) 3*s* B) 4*s* C) 3*d* D) 3*p*

6. An object is placed in front of the concave mirror then object distance, image distance and magnifications as per the sign conventions i.e. $-u, +v, +m$ for the following one diagram is ( )

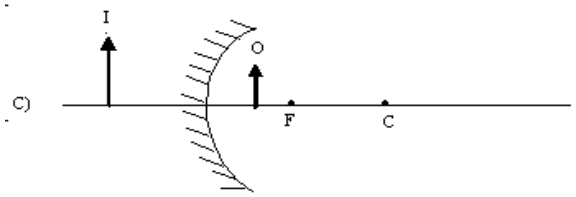
A)



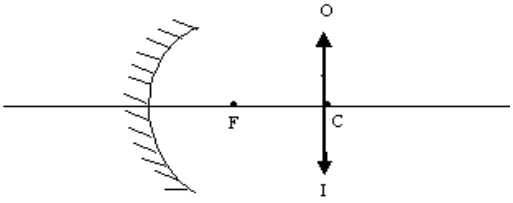

B)




C)




D)

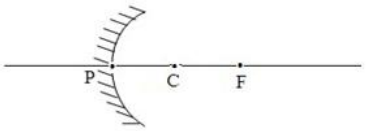



7. Number of orbitals present in the sub shell if n = 4 ( )

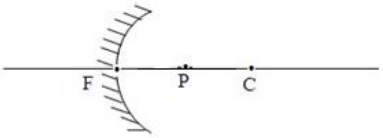
A)16 B) 9 C) 4 D) 10

8. Correct ray diagram from the following is ( )

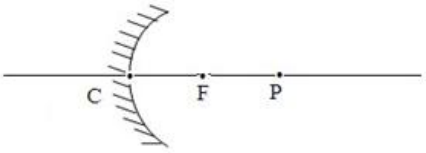
A)



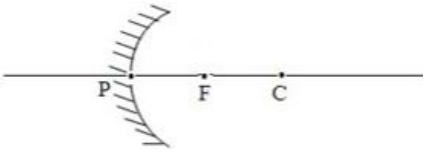

B)




C)

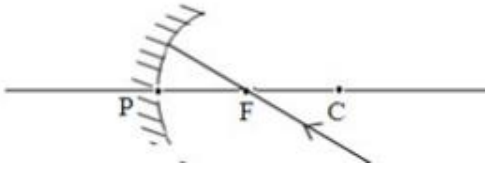



D)

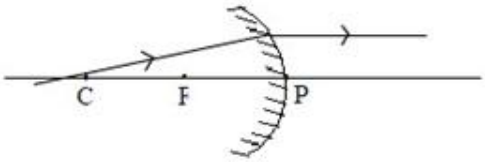



9. Correct ray diagram from the following is ( )

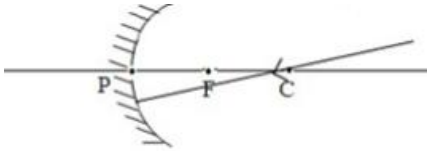
A)



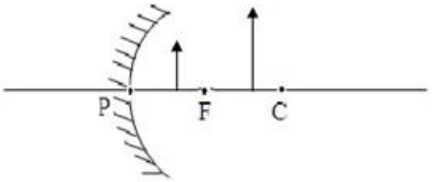

B)




C)




D)




10. Image formed when we read the paper by using magnifying glass is ( )

A) Virtual image B) real image

C) Image formed at infinity D) smaller than object.